

Speaker script

Network Position and Academic Performance

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ECC3841 Network Economics — Presentation 2 — Progress Report 10 minutes

Opening

about 0.4 min

Good morning. I'm Tingying Huang, presenting joint work with Amy Bing on network position and academic performance. Last time we asked whether a student's position in their friendship network predicts their grades, on top of who their direct friends are. Today is about what actually happened when we ran that test — including a gap in our own design that we found and had to fix.

Live cue: Greeting; coauthor; recap the question in one sentence; frame today as results, not setup.

Does network position predict grades?

about 0.9 min

Quick recap of where the question comes from. Smirnov and Thurner showed that friends' grades move together mostly because students pick friends with similar GPA. It's not that friends pull each other up or down. But their method only looks at one number, a friend's average GPA, and it ignores the rest of the network. Calvó-Armengol, Patacchini and Zenou showed that network position itself, measured with Katz-Bonacich centrality, predicts achievement. But their data is a single snapshot. So they can't tell whether position raises grades, or good grades just attract more friends. Our plan combines the two: take the centrality prediction to Smirnov and Thurner's panel data, and test it while holding friend-group GPA and a student's own past GPA fixed.

Live cue: Two antecedents in one sentence each; the gap each leaves; our plan combines them.

Recap: design and data

about 0.8 min

Here's the regression we planned. Next term's GPA depends on Katz-Bonacich centrality, on the student's own past GPA, on their friends' average GPA, and on raw in- and out-degree, plus group fixed effects. Friend GPA nets out friend-group composition. Degree nets out raw popularity. And we sweep phi from near zero, where centrality is basically popularity, to near its ceiling, where it captures long reach through the network. One fact about the data matters a lot for what comes later. GPA is measured six times over for the school group. But for each of the four university cohorts, it's a single static number, not something that varies over time.

Live cue: Regression equation; each control's job; phi dial; the panel-vs-static data fact, stated plainly.

What we did

about 0.5 min

Four steps. Replicate Smirnov and Thurner's own checks, to confirm the data behaves the way they reported. Run centrality on its own, with no popularity control — the naive test. Add popularity, sweep phi, and pool all five groups — the test we planned last time. And check the assumptions behind that test — where we found one had failed. Here's what actually happened at each step.

Live cue: Four steps, fast; land on "here's what happened."

The replication holds

about 0.8 min

First check: does a student's own past GPA predict their next GPA? Yes, very strongly — this is by far the largest coefficient in the whole regression. Does a friend's past GPA add anything on top of that? No — not once you control for the student's own trajectory. Second check: are new friendships more GPA-similar than friendships about to end? In the university data, yes, clearly. In school, the same direction, but not significant, probably because the school sample is smaller. Both of these match Smirnov and Thurner's original findings, now in this new panel setting with a real time lag. That's the foundation everything else in this talk builds on.

Live cue: Own-GPA strong, friend-GPA null; new-vs-dropped; "matches prior work" as foundation.

The naive effect is popularity, relabelled

about 0.8 min

Now the first genuinely new test. Katz-Bonacich centrality on its own, net of friend GPA, pooled across all five groups: positive, and highly significant. Looks like position matters. But add raw in-degree — just how many people like you — to that same regression, and the centrality coefficient drops to essentially zero, no longer significant. Meanwhile in-degree itself is large and highly significant. That's the tell: centrality and popularity are highly correlated in these networks, so whichever one you leave out of the regression looks like it's doing all the work. So the naive centrality effect wasn't really about network position at all. It was popularity, relabelled.

Live cue: Naive positive; add degree, it vanishes; degree itself is what was doing the work.

The headline test

about 0.5 min

This is the test we actually planned last time. Popularity and friend GPA are both controlled. Phi is swept across six values, pooled across all five groups. At phi equals 0.85 — the value we fixed in advance as our headline — we get a small, negative, statistically significant coefficient. Read at face value, that says network position has a small, independent, negative effect on grades, net of popularity and friend quality.

Live cue: The sweep table briefly; the fixed headline number; "read at face value" signals more is coming.

But we found a gap in it

about 0.6 min

Here's the problem we found. That regression was supposed to control for a student's own past GPA. But remember: GPA is only measured repeatedly for the school group. For all four university cohorts, it's a single static number in the data, not something that varies over time. You literally cannot build an own-past-GPA regressor for over eighty percent of that pooled sample. The model we promised last time had that control written into it. The test we actually ran did not have it.

Live cue: Restate the promised control; the static-GPA fact; the honest admission, in the brief's own words.

The corrected test

about 1.4 min

So we restricted the confirmatory test to the one valid sample. That's the school group, with a real time lag and a real own-past-GPA control. It comes to two thousand and eighty-eight observations, from five hundred thirty-five students. Before trusting it, we checked whether that own-GPA control collides with the other regressors. It doesn't. The correlations are all small, and the variance inflation factor for own-GPA is 1.11 — textbook territory. Katz and in-degree still correlate highly with each other, around 0.87, but that's expected, and it's exactly what sweeping phi across the whole range is designed to handle. The result: at every single value of phi, the coefficient is not significant, and the sign isn't even consistent from one end of the dial to the other. Put the three tests side by side and you get three different pictures. The naive test, positive. The pooled test, negative. The corrected test, looking like nothing at all. Net of popularity, friend GPA, and a student's own past performance, once the test is actually specified correctly, we find no evidence that network position independently predicts grades.

Live cue: Sample and validity; the VIF check briefly; all six phi values null; the three-way picture; the headline.

Conclusion

about 0.9 min

So here's the answer. Net of popularity, friend GPA, and a student's own past performance, there's no evidence that network position predicts grades on its own. Why does that still matter? Because the progression itself is the finding — a naive test that looked positive, a pooled test that looked negative, and a correctly specified test that looks like nothing, all from the same data. Pooling groups with mismatched data structures can manufacture a significant result that the one correctly specified subsample simply doesn't support. Practically, that means interventions built around widening students' network reach aren't supported by what we've found here. Next, we're testing whether that null result comes from weak, one-way "likes," by looking only at mutually confirmed ties.

Live cue: Headline answer; the progression as the real finding; implication; one-line forward pointer. No "thank you."

Q&A appendix

conditional; excluded from talk time

Sample and network construction

if asked

Give the Harvard Dataverse source and CC0 licence, the roughly 6,200 Russian students, the node counts by cohort, and the directed “like” tie definition with its 3-month windows. If asked why “likes” rather than self-reported friendship: name the 24–27% reciprocity rate directly, and connect it to the reciprocal-ties test planned next.

Robustness: the full ϕ range

if asked

Walk through the two lines on the chart. Before the own-GPA fix, pooled swings from significantly positive at low phi to significantly negative at high phi. After the fix, school-only stays flat and non-significant across the same range. The point isn’t that one phi value is right — it’s that the corrected line never leaves the null region anywhere on the dial.

Checking for multicollinearity

if asked

If someone worries that adding a student’s own past GPA should collide with the other right-hand-side variables: walk through the correlation table — 0.008 with Katz, 0.065 with in-degree, 0.122 with out-degree, 0.285 with friend GPA — and the VIF of 1.11. That correlation with the outcome is the whole point of a control variable; what would be a problem is correlation with the *other regressors*, and that’s what this table rules out.

What’s next: reciprocal ties

if asked

Only 24 to 27 percent of “like” ties go both ways. The next step recomputes centrality using only those mutually confirmed ties. That tests directly whether a one-way “like” is too weak a tie for the model’s mechanism to operate through, and whether the null result changes once ties are restricted to ones both people confirm.

Related literature

if asked

Calvó-Armengol, Patacchini and Zenou (2009) is the model itself, tested on a single cross-section. Smirnov and Thurner (2017) is the paper and data we extend. Giulietti, Vlassopoulos and Zenou (2020) is a different peer effects design, using reference groups rather than friendship ties, by the same course instructor. Positioning: this is the first time centrality has been tested with a real time lag and an own-performance control, which the original cross-sectional data could not support.

Standing Q&A stance

Hear the whole question and count to three. Answer its strongest version in one sentence before adding detail. If asked why we're presenting a null result: say plainly that the progression — naive, pooled, corrected — is itself the finding, and that the corrected test is the only honest one.

Live cue layer

1. **Does network position predict grades?** — two antecedents, the gap each leaves, our combined plan.
2. **Recap: design and data** — the equation, each control's job, the static-GPA fact.
3. **What we did** — four steps, fast.
4. **The replication holds** — own-GPA strong, friend-GPA null, new-vs-dropped.
5. **The naive effect is popularity, relabelled** — positive, then vanishes with degree.
6. **The headline test** — the sweep, the fixed 0.85 headline, "at face value."
7. **But we found a gap in it** — the promised control that wasn't actually there.
8. **The corrected test** — valid sample, VIF check, all-phi null, the three-way picture.
9. **Conclusion** — the answer, the progression as the finding, the forward pointer.